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Palaeomagnetism of two Mid-Tertiary Basaltic Volcanoes in Queensland Australia

P. WELLMAN *

Research School of Earth Sciences
Australian National University, Canberra, A.C.T.

SYNOPSIS

A palaeomagnetic investigation of 77 flows from lava successions in the Tweed Volcano and Main Range Province provides stable directions from 63 flows after magnetic cleaning. 7 flows are regarded as being intermediate, arising from sampling polarity transitions or geomagnetic excursions. The remaining 56 flows give a palaeomagnetic pole at 114.8° E, 74.2° S with A_{95} = 4.1°. K-Ar dating restricts the vulcanicity to the period 24-20 m.y. (Late Oligocene to Early Miocene).

INTRODUCTION

Cenozoic mafic volcanism is widespread along the eastern margin of Australia. A regional K-Ar isotopic dating study of these rocks (Wellman and McDougall, 1974a) has shown that the volcanic activity ranges from late Cretaceous to Recent. The volcanic rocks are mainly basaltic lavas forming squat shield volcanies or thin lava fields. Deep dissection has exposed thick lava sections. Palaeomagnetic studies have been carried out on lava sections in nine volcanies, the results from three volcanies in northern New South Wales are reported in Wellman *et al.* (1969), the results from two volcanoes are reported here, and the results from the four other volcanoes and a synthesis of all Cenozoic results from Australia are given in McElhinny *et al.* (in press). The palaeomagnetic technique used in all these studies is described in Wellman *et al.* (1969) and will not be repeated here.

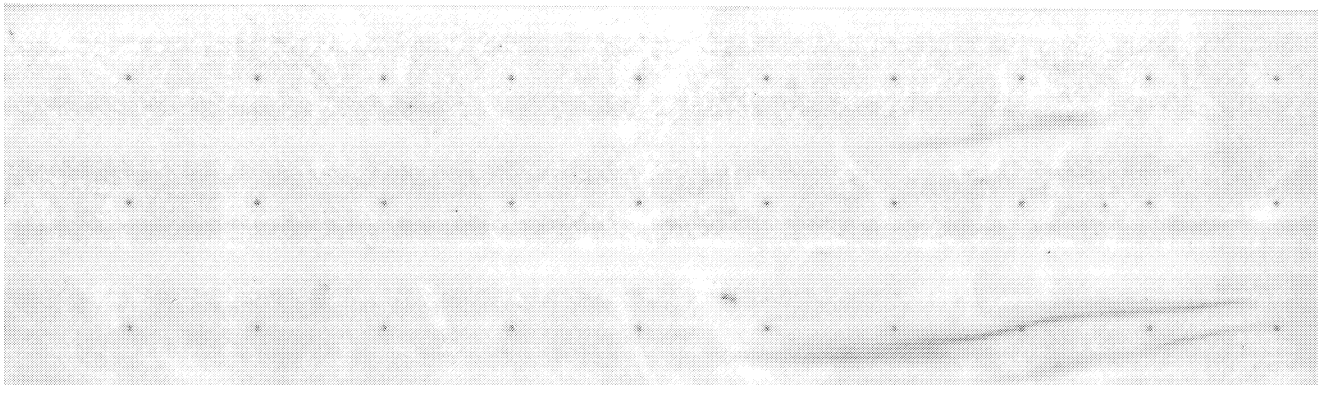
GEOLOGY AND SAMPLING

TWEED VOLCANO

Lying astride the N.S.W. - Queensland border is the Tweed Volcano — a large shield volcano composed mainly of basaltic flows and approximately 100 km in diameter and 1000 m maximum thickness. The lavas are mostly moderately over-saturated tholeiites, and range from basalts to rhyolites, with minor transitional and possibly alkaline basaltic rocks and some alkaline trachytes and alkaline rhyolites (Richards, 1916; Wilkinson, 1968).

The section sampled for palaeomagnetic study is near Mount Hobwee, on the northern margin of the central erosional caldera (28.2° S, 153.2° E). The upper part of the section was sampled on a spur to the southwest of Mount Merino (Tyalgum 1:50,000 map, grid reference 218733 to 202745), while the lower part (below 620 m above sealevel) was sampled in Nerang Creek and its tributary ENE of Mount Merino (Canungra 1:50,000 map, grid reference 228791, through 229756 to 209751). The combined section is 940 m thick but only about one third is exposed, the flows averaging about 6 m thick. In this section a felsic horizon crops out at 440 m (? Hillview Rhyolite) and the Binna Burra Rhyolite of McTaggart (1962) extends from 608 to 651 m. Three samples from near the base of the section give whole rock K-Ar ages (Wellman and McDougall, 1974b) of 22.4 ± 0.9 m.y. at 312 m, 22.2 ± 0.9 m.y. at 330 m and 22.2

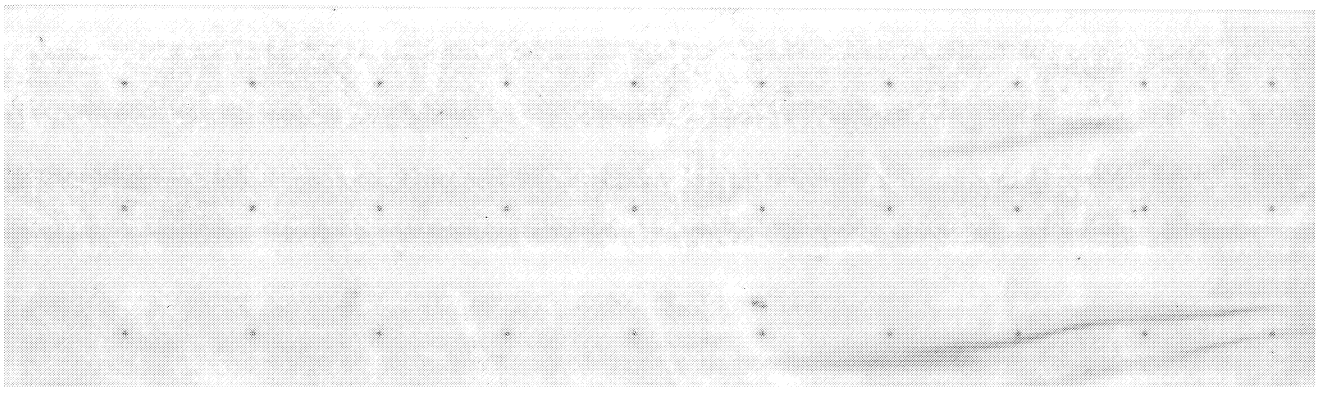
* Present address: Bureau of Mineral Resources, Geology and Geophysics, Canberra, Australia.



± 0.9 m.y. at 440 m, two glassy samples from the Binna Burra Rhyolite give whole rock ages of 12.7 ± 0.5 and 20.8 ± 0.8 m.y., and two samples from near the top of the section give whole rock ages of 20.1 ± 0.8 m.y. at 824 m and 20.0 ± 0.8 m.y. at 914 m. It seems likely that the sample giving the age of 12.7 m.y. has lost argon by diffusion, and that the section of lavas was erupted between 22.3 ± 0.6 and 20.0 ± 0.6 m.y. ago (Early Miocene). There is no evidence for significant tilting or faulting of the flows in the sampled area. McTaggart (1961) mentions dips of up to 2° in the part of the shield to the west of Mount Hobwee, and shows that the overall tilting of the volcano is less than one degree. One hundred and one samples were collected from 52 flows in the volcanic succession.

Table 1: Mean Flow Direction of Magnetization after magnetic cleaning, Tweed Volcano — Mt. Hobwee

Flow No.	Altitude (m)	D (deg.)	I (deg.)	Polarity
42	914	197.0	+64.4	R
41	902	188.2	+69.4	R
40	898	193.4	+54.7	R
39	883	181.5	+49.7	R
38	867	177.5	+57.9	R
37	848	177.4	+76.7	R
36	824	180.5	+56.4	R
35	707	163.0	+64.0	R
34	685	181.2	+59.7	R
33	675	165.9	+63.3	R
32	666	168.6	+62.9	R
31	629	200.3	+56.7	R
30	624	200.0	+57.0	R
29	595	232.8	+72.5	R
28	589	004.4	−52.4	N
27	578	001.6	−49.1	N
26	575	010.4	−54.6	N
25	572	003.3	−54.9	N
24	566	001.5	−52.5	N
23	561	003.6	−52.9	N
22	556	012.3	−50.5	N
21	545	348.8	−71.8	N
20	519	352.8	−59.5	N
19	506	005.2	−53.3	N
18	499	014.5	−44.0	N
17	485	026.4	−46.2	N
16	465	000.6	−24.6	
15	462	351.6	−20.2	
14	454	344.6	−22.5	
13	432	351.5	−16.7	
12	419	350.6	−24.8	
11	413	199.1	+71.6	R
10	409	241.0	+70.1	R
9	405	243.0	+64.1	R
8	330	200.5	+52.3	R
7	324	232.6	+68.1	R
6	312	195.4	+60.6	R
5	300	358.0	−72.0	N
4	296	011.9	−64.2	N
3	291	019.2	−62.0	N
2	249	184.4	+60.2	R
1	205	216.3	+42.6	R



Main Range Province

A thick pile of volcanic rocks known as the Main Range Volcanics (Stevens, 1965) occurs about 100 km south-west of Brisbane on the Main Dividing Range near the southern margin of Queensland. The volcanics form a lava pile about 70 km long, 40 km wide and 900 m thick. They consist mainly of basaltic flows, but they range in composition from olivine basalt to comendite, and there is also considerable trachyte breccia (Richards, 1916).

The lava section sampled is in the eastern central part of the province (28.2° S, 152.4° E;, and consists of a road-cutting on the Cunningham Highway east of Cunninghams Gap (Warwick 1:100,000 map, grid reference 410964 to 402969), and stream and cliff exposures southeast of Mount Cordeaux (grid reference 396970 to 402978). This section consists mainly of basaltic lavas, with minor trachyte below 757 m above sealevel (Stevens, 1965). Four K-Ar whole rock ages have been measured by Webb *et al.* (1967) in the area studied; samples from near the base of the section gave ages of 23.9 ± 0.8 m.y. at 534 m and 23.7 ± 1.0 m.y. at 695 m, and samples from near the top of the section on the nearby Mount Mitchell gave ages of 22.3 ± 1.0 m.y. at 1015 m and 22.1 ± 0.8 m.y. at 1142 m. The volcanic activity therefore took place in the Late Oligocene. There is no evidence for strong tilting or faulting of the flows. Stevens (1965) states that the lavas and pyroclastic rocks dip easterly and westerly at low angles (1.5° - 4°) away from a NNW-SSE line close to the present divide. He thought that these dips were primary and that the lavas were extruded from fissures along the line of the main divide. Forty-six samples were collected from 25 flows in this section. Two of these flows were also investigated by Robertson (1966). In addition a rhyolite sample was collected from the 23.3 ± 1.8 m.y. old (Webb *et al.*, 1967) Mount Alford ring complex.

PALAEOMAGNETIC MEASUREMENTS

Tweed Volcano

Samples varied widely in magnetic properties, the basaltic flows generally having initial intensities of magnetization between 30 and 5 × 10^{-4} emu. cm^{-3} whilst the felsic flows had relatively low intensities ranging from 400 to 1 × 10^{-6} emu. cm^{-3}. Thirteen samples were classified as having unstable directions of magnetization, and for four flows the directions of the two samples differed by more than 25° and were

Table 2: Mean Flow Direction of Magnetization after magnetic cleaning Main Range Province — Mount Cordeaux.

Flow No.	Altitude (m)	D (deg.)	I (deg.)	Polarity
21	1067	009.9	−55.8	N
20	1013	275.4	−86.7	
19	1002	145.9	−68.9	
18	994	335.8	−53.7	N
17	980	042.4	−70.7	N
16	948	037.5	−44.0	N
15	885	108.6	+79.1	
14	857	194.6	+53.6	R
13	818	014.4	−53.7	N
12	811	017.7	−51.5	N
11	805	004.4	−47.5	N
10	783	012.5	−58.9	N
9	770	020.2	−56.5	N
8	754	030.0	−56.5	N
7	748	015.7	−35.4	N
6	743	359.9	−44.1	N
5	719	337.0	−38.9	N
4	751	021.0	−50.6	N
3	742	008.0	−62.8	N
2	739	002.6	−58.0	N
1	735	220.5	+57.2	R
	Intrusion	010.2	−45.8	N

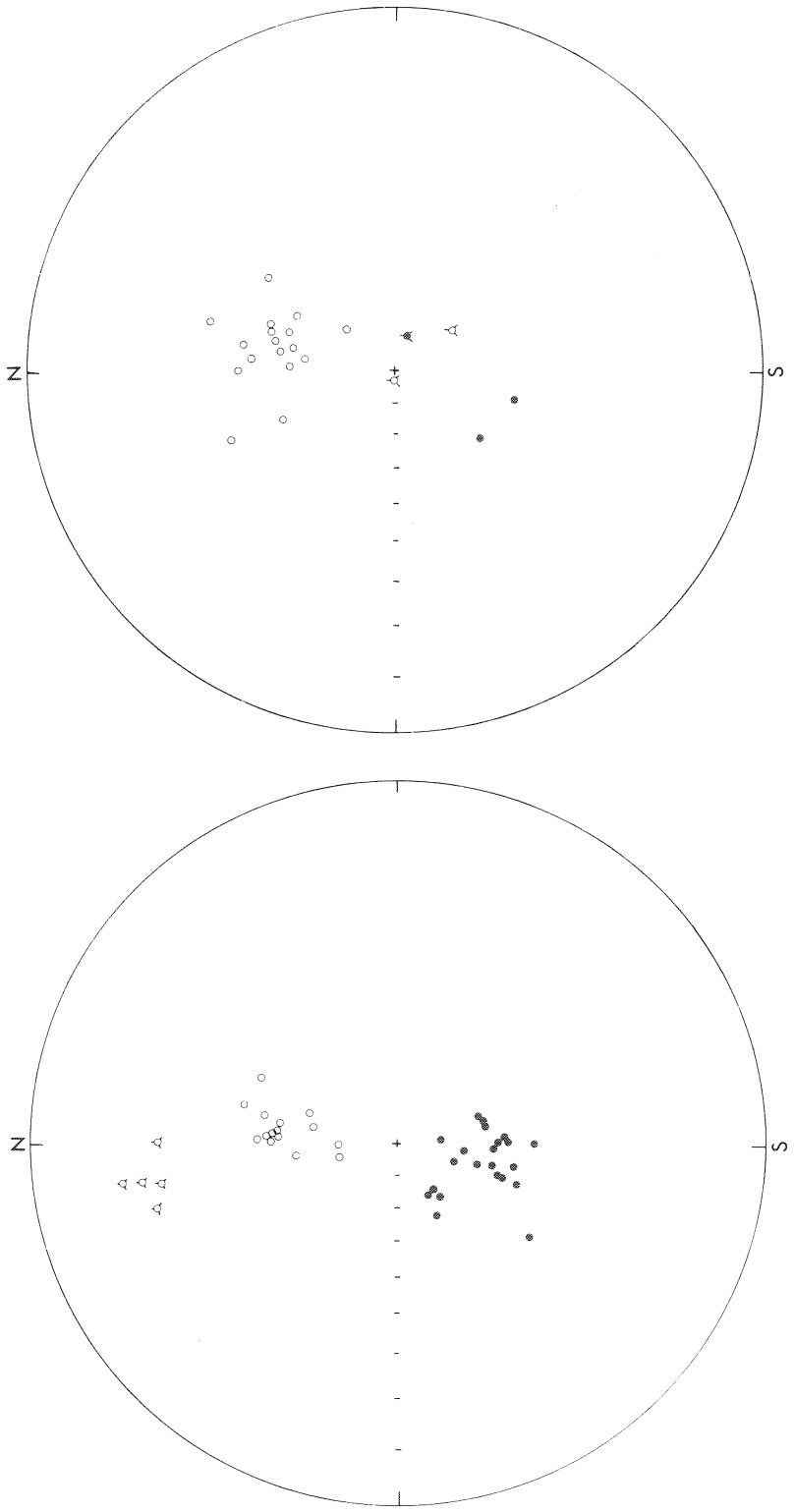


FIG. 1: Stereographic projection of stable directions of magnetization from lava flows in the (a) Tweed Volcano and (b) Main Range Province. Solid circles on lower hemisphere, open circles on the upper hemisphere. Directions regarded as intermediate are starred.

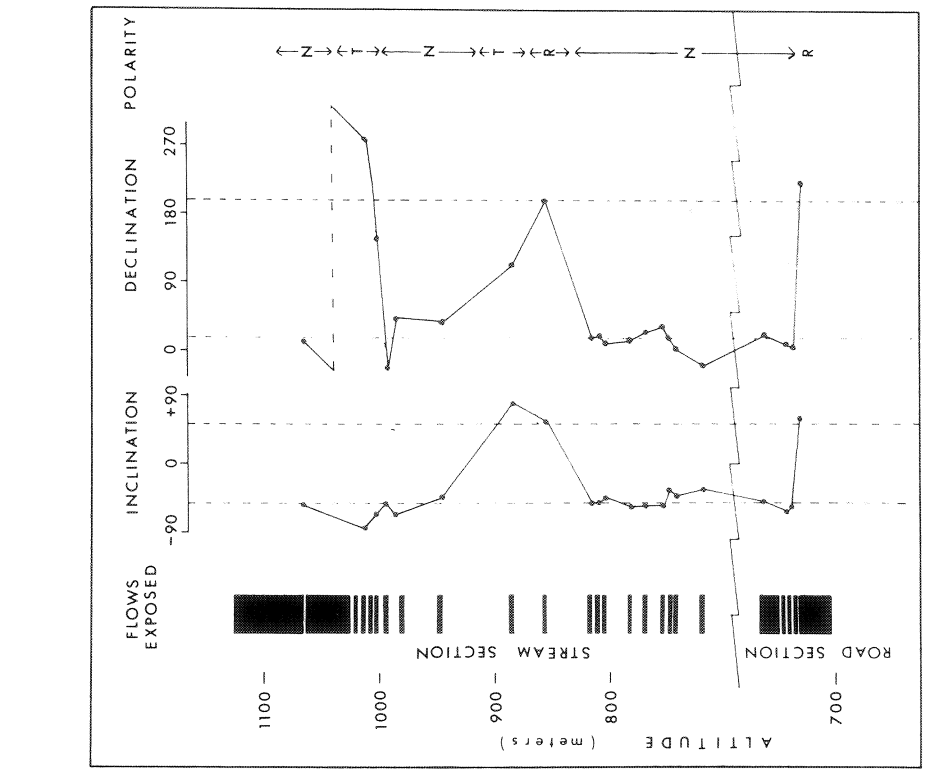


FIG. 3: Main Range Province. Variations in magnetic declination and inclination from the overall mean direction (dashed lines).

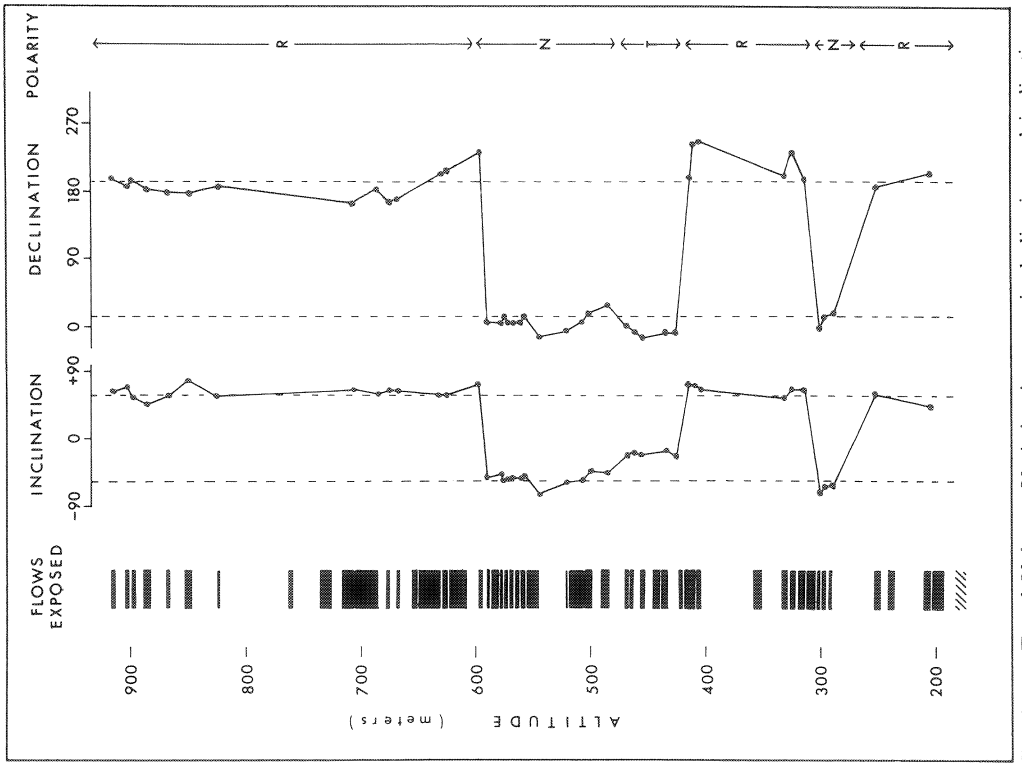


FIG. 2: Tweed Volcano. Variation in magnetic declination and inclination, from the overall mean direction (dashed lines).

therefore rejected. The directions of magnetization of the remaining 42 flows after magnetic cleaning are listed in Table 1 and shown in Fig. 1, a. The section is interpreted (Fig. 2) as showing four changes in polarity, with five flows falling in the transition zone in one case. These five flows are not included in the overall mean, calculated from 37 flows, in Table 3.

The K-Ar dating indicates that the period of volcanism in the Tweed Volcano was 2.2 ± 0.9 m.y. (22.3 ± 0.6 m.y. to 20.1 ± 0.6 m.y.). This period is longer than for other volcanoes of this type, so there is a possibility of a major time break in the volcanic sequence. The most likely place for a break is at the base of the Binna Burra Rhyolite. At this horizon there is a change in magnetic polarity with no transitional flows, there seems to be a change to more tholeiitic composition, and there is an apparent change in age from 22.4 ± 0.9 − 22.2 ± 0.9 m.y. to 20.8 ± 0.8 − 20.0 ± 0.8 m.y.

MAIN RANGE PROVINCE

Intensities of magnetization of basaltic flows ranged between 20 and 6 × 10^{-4} emu. cm^{-3}, whilst the rhyolite had an initial intensity of 6 × 10^{-6} emu. cm^{-3}. Demagnetization procedures showed that six samples were to be considered as unstable. Two samples from one flow differed by more than 25° after magnetic cleaning and were rejected. The directions of magnetization of the remaining 22 flows and one intrusion after magnetic cleaning are given in Table 2 and illustrated in Fig. 1, b. The sequence is interpreted (Fig. 3) as indicating three polarity changes, including one transitional direction, and also an excursion represented by two flows not related to a polarity change. These three flows were omitted from the overall mean given in Table 3.

Sites F and G of Robertson (1966) correspond in position to flows 19 - 20 and 3 - 4 of Table 2, and the magnetization directions are similar.

Table 3: Overall results from Tweed Volcano and Main Range Provinces.

	Age (m.y.)	N	D	I	R	K	α_{95}
Tweed Volcano	22 - 20	37	192.5	+60.5	36.15	42.4	3.7
Main Range	24 - 22	19	192.3	+53.6	18.50	35.9	5.7
Combined	24 - 20	56	192.4	+58.2	54.56	38.2	3.1

N = number of flows used in calculating mean, D = declination clockwise from true north, I = inclination positive downward, R = resultant length of N unit vectors, K = precision parameter, α_{95} = radius of circle of 95 percent confidence.

OVERALL RESULT

Table 3 gives the mean directions for the Tweed and Main Range sequences separately and also a combined result for the 56 flows. This provides a fairly well defined pole position for Australia for the time range 24 - 20 m.y. (Late Oligocene to Early Miocene) at 114.8° E, 74.2° S with 95 percent circle of confidence of 4.1°.

The results from these lava sequences can be combined with the results of Robertson (1966) to obtain a slightly more reliable pole position for 25 - 20 m.y. (Lat. 77.4° S, Long. 110.9° E, α_{95} = 4.3°; McElhinny *et al.* 1974). This pole is intermediate in latitude to the 35 - 25 m.y. and the 20 - 15 m.y. poles of McElhinny *et al.* (1974), although rather unexpectedly it lies well to the east of both poles. The southeast Queensland palaeomagnetic results therefore help to confirm the systematic decrease in the palaeolatitude of Australia between 95 and 20 - 15 m.y. ago, but they suggest that during this period Australia may have rotated several times relative to the pole.

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